Three-Phase Induction Motor Torque-Speed & Equivalent Circuit

Rotating magnetic field theory, per-phase equivalent circuit derivation, maximum breakdown torque formulas, and circle diagram analysis.

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1. Rotating Magnetic Field & Slip

Three-phase stator winding currents displaced by 120 deg establish a constant magnitude flux rotating at synchronous speed Ns = 120*f/P.

2. Torque-Slip Characteristic

Torque is proportional to s*E2^2*R2 / (R2^2 + (s*X2)^2). Maximum breakdown torque occurs at slip s_max = R2 / X2.